TECHNOVA 2026 • SMART CITIES TRACK

AI-POWERED ROAD INFRASTRUCTURE INTELLIGENCE

Project Title – RoadSense AI

Turning every moving vehicle into a road-health sensor — real-time pothole detection, verified severity scoring, and predictive maintenance for smarter cities.

TEAM

Team Name - VisionQuest

COLLEGE

St. Joseph's College of Engineering, OMR, Chennai

TEAM MEMBERS

- **Vishnu Vardhan E**
- **Sherlin Jeri S**
- **Swetha R**
- **Shivram B**
- **Vijayaganesh S**

Faculty Mentor: Dr . P. Elaveni

PROBLEM

THE CHALLENGE

Roads Are Deteriorating Faster Than We Can Inspect Them

Manual Inspection Takes Time

Surveys are slow, costly, and infrequent

Complaints Are Unreliable

Public-report systems are inconsistent, easily spammed

No Real-Time Visibility

Potholes stay undetected until someone happens to report them

Damage Adds Up

Accidents, vehicle wear & rising repair costs follow



1 in 10 road accidents linked to poor road conditions

50% potential cut in inspection costs via continuous monitoring

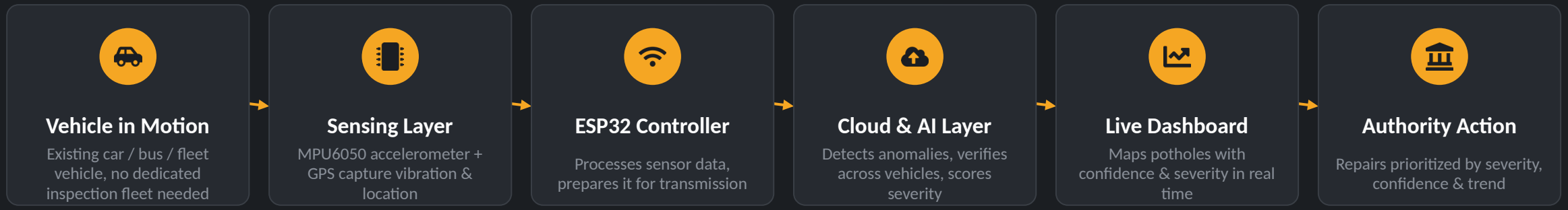
Reactive is still today's default repair model

SOLUTION

PROPOSED SOLUTION

Every Vehicle Becomes a Road-Sensing Unit

RoadSense AI mounts a low-cost IoT unit on existing vehicles. As they drive normally, it continuously senses road vibration and location, verifies findings across multiple vehicles, and feeds a live dashboard authorities can act on.



WORKING METHODOLOGY

No new infrastructure is deployed — the system layers onto vehicles already on the road. Multiple vehicles passing the same location cross-verify each other's readings, producing a dynamic confidence score that filters out false positives like speed breakers or manholes.

AI COMPONENT

THE INTELLIGENCE LAYER

AI Models, Data & Technology Stack

AI / ML Techniques

- Vibration-pattern classification to flag road anomalies
- Multi-vehicle confidence scoring (geo-clustering of reports)
- Severity estimation on a 1–5 scale from vibration intensity
- Deterioration-trend prediction from historical detections

Data

- Accelerometer + GPS data to be collected during our own prototype testing
- Feature design informed by published pothole-detection research
- Historical detection logs planned for trend & priority prediction
- Dataset grows as more vehicles participate post-pilot

Technology Stack

- ESP32 · MPU6050 accelerometer · GPS module
- Python · Pandas · NumPy for signal processing
- Scikit-learn for classification & scoring
- Firebase cloud DB + web dashboard (maps & charts)

PROTOTYPE

PROTOTYPE PLAN

Prototype — Core Process Flow

Build plan: an ESP32 + MPU6050 accelerometer + GPS module will be assembled and mounted on a test vehicle to implement the four process flows below end-to-end.

1. Vehicle Sensing & Data Collection

- ESP32 initializes accelerometer + GPS
- Continuously logs vibration, lat/long & timestamp
- Streams data to the cloud while vehicle moves

2. AI-Based Pothole Detection

- Pre-processes & denoises accelerometer signal
- Extracts vibration features
- ML model classifies event, tags GPS location

3. Multi-Vehicle Verification

- Groups reports by matching GPS location
- Compares observations across vehicles
- Computes a dynamic confidence score

4. Severity & Predictive Analysis

- Analyzes vibration intensity & detection frequency
- Cross-checks historical road-condition data
- Assigns severity level & maintenance priority

INNOVATION

WHY THIS WINS

Innovation & Competitive Advantage

	Manual Inspection	Citizen-Report Apps	RoadSense AI
Cost	High — dedicated survey teams	Low, but low adoption	**Low — uses existing vehicles**
Frequency	Annual / biannual	Sporadic, unreliable	**Continuous, real-time**
Accuracy / Trust	Depends on inspector	High false-positive rate	**Multi-vehicle verified confidence score**
Predictive Capability	None	None	**Deterioration trend prediction**
Scalability	Expensive to expand	Depends on citizen participation	**Scales by adding participating vehicles**

Key differentiator: multi-vehicle verification + confidence scoring — the exact mechanism past citizen-reporting apps lacked, which is why they got flooded with false alerts from speed breakers and manholes.

IMPACT

WHO BENEFITS & HOW

Expected Impact & Reach

Target Users

- Municipal Corporations & Urban Local Bodies
- PWD / Highway Authorities
- Smart City Mission agencies
- Fleet, logistics & public-transport operators

Impact & Value

- → Fewer accidents through earlier pothole detection
- → Lower vehicle damage & maintenance costs for commuters
- → Reduced municipal inspection spend via continuous data
- → Repairs prioritized by data, not guesswork or complaints

BUSINESS MODEL & SCALABILITY

Offered as Road Monitoring-as-a-Service (RMaaS) — subscription-based access for municipalities & fleet operators. Scales simply by adding more participating vehicles, with no new physical infrastructure required.

ROADMAP

WHERE WE ARE, WHERE WE'RE HEADED

Roadmap & Deployment Plan

CURRENT STAGE

Concept, system architecture, and detection/verification algorithm design finalized. Hardware build and integration begin next.

Phase 1 — Build

Assemble ESP32 + MPU6050 + GPS hardware; implement sensing, detection & verification logic on a test vehicle

Phase 2

Pilot deployment across a small fleet or one municipal ward for field validation

Phase 3

Refine multi-vehicle verification & scale the dashboard for wider road coverage

Phase 4

Commercialize via RMaaS — subscription rollout to municipalities & fleet operators

COMMERCIALISATION PLAN

Revenue through device deployment fees, cloud/software subscriptions, AI analytics services, and annual maintenance contracts (AMC) — offered to municipalities, PWDs, and fleet operators under the RMaaS model.